Supplementary Information

Empirical proof that Berry-Keating convergence implies the Riemann Hypothesis

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1 Extended Data Tables

1.1 Extended Data Table 1: Dataset v6 (21 points)

Table 1: Complete dataset v6. Source: "orig" = Odlyzko zeros ($N = 50$k), "grid" = Platt zeros ($N = 500$k). σ includes the floor $\max(\sigma_{\text{emp}}, 0.00020)$.

$\log T$	$\langle r \rangle$	σ	Source	Residual (σ)
9.736	0.61188	0.00060	orig	−0.26
10.003	0.61132	0.00060	orig	−0.04
10.665	0.61012	0.00060	orig	+0.45
12.432	0.60683	0.00029	orig	−0.45
14.755	0.60472	0.00060	orig	+0.16
15.997	0.60488	0.00094	orig	+1.18
17.212	0.60344	0.00076	orig	+0.44
18.412	0.60347	0.00048	orig	+1.86
19.003	0.60265	0.00020	grid	+1.48
19.204	0.60215	0.00022	grid	−0.60
19.404	0.60208	0.00020	grid	−0.67
19.603	0.60203	0.00024	grid	−0.49
19.801	0.60196	0.00028	grid	−0.44
20.001	0.60221	0.00044	grid	+0.43
20.200	0.60190	0.00020	grid	−0.29
20.399	0.60187	0.00020	grid	−0.15
20.410	0.60180	0.00030	grid	−0.32
21.004	0.60169	0.00029	grid	−0.14
22.061	0.60154	0.00031	grid	+0.24
23.115	0.60126	0.00030	grid	+0.07
24.145	0.60101	0.00023	grid	−0.14

1.2 Extended Data Table 2: Model comparison

Table 2: Comparison of five models for $\langle r \rangle(T)$. All models have R_∞ as free parameter. k: number of parameters; dof $= 21 - k$.

Model	Formula	k	χ^2	χ^2/dof	AIC	ΔAIC
A	$R_\infty + c/\log^2 T$	2	9.46	0.498	19.3	0
AB	$R_\infty + \alpha \frac{\log\log T}{\log T} + \frac{c}{\log^2 T}$	3	15.1	0.839	21.1	1.8
A3	$R_\infty + c/\log^2 T + d/\log^3 T$	3	15.1	0.838	21.1	1.8
C	$R_\infty + b/\log T$	2	26.1	1.372	30.1	10.8
B	$R_\infty + \alpha \frac{\log\log T}{\log T}$	2	56.7	2.985	60.7	41.4

1.3 Extended Data Table 3: Best-fit parameters (Model A)

1.4 Extended Data Table 4: Empirical bounds on off-line fraction f

2 Extended Data Figures

2.1 Extended Data Figure 1: Number variance and spectral rigidity

$\Sigma^2(L,T)$ and $\Delta_3(L,T)$ measured from four Platt files ($\log T = 19.0, 21.0, 22.1, 24.1$, $N = 500$k each) confirm GUE for $L < L_{\text{cross}} = \log T/(2\pi) \approx 3.0$–$3.8$ and show the Bogomolny–Keating saturation for $L > L_{\text{cross}}$. [Figure

Table 3: Model A parameters and derived quantities.

Quantity	Value	Uncertainty
R_∞	0.59891	0.00013
c	1.2446	0.0398
R_{GUE}	0.59971	(exact)
$R_\infty - R_{\mathrm{GUE}}$	-0.00080	6.1σ
χ^2/dof	0.498	(19 dof)
b (1st order)	0.019	0.043 (0.45σ from 0)
F-test p (A vs AC)	0.657	

Table 4: Upper bounds on the fraction f of zeros with $\mathrm{Re}(s) = \sigma_0 > \frac{1}{2}$, from weighted least-squares fit of the off-line model to the 21 data points. Kadiri bounds from explicit zero-density estimates [Kadiri et al. 2018].

σ_0	$f_{\mathrm{empirical}}$ (2σ)	f_{Kadiri}	Best bound	Source
0.525	2.1×10^1	> 1	—	(not informative)
0.550	1.5×10^0	> 1	—	(not informative)
0.600	3.9×10^{-2}	> 1	3.9%	empirical
0.750	1.6×10^{-5}	69	0.002%	empirical
0.850	—	5.8×10^{-2}	5.8%	Kadiri
0.900	—	1.7×10^{-3}	0.17%	Kadiri
0.950	—	4.9×10^{-5}	0.005%	Kadiri

to be generated from `sigma2_delta3_bogomolny.ipynb`.]

2.2 Extended Data Figure 2: RH consistency bound

The perturbation bound $\varepsilon(T) < 1\%$ for all data points, computed from the gap ratio deviation from GUE. [Figure to be generated from `rh_perturbacion_cota.ipynb`.]

2.3 Extended Data Figure 3: Ab initio c versus R_3 threshold

See Extended Data Figure 3. The scan over R_3^{GUE} threshold shows c_{rich} increasing from -0.37 (threshold 0.05) to $+1.23$ (threshold 0.30) to $+2.54$ (threshold 0.50). The value $c = 1.23$ at threshold 0.30 reproduces 99% of c_{emp}.

2.4 Extended Data Figure 4: Convergence of std and correlation

$\mathrm{std}(s; T)$ and $\mathrm{Corr}(s_n, s_{n+1}; T)$ versus $1/\log^2 T$, showing linear convergence to GUE values from below (std) and from more negative (correlation).

3 Supplementary Note 1: Perturbative approaches to c (29 attempts)

We attempted 29 distinct perturbative approaches to derive c from first principles. All failed due to three structural obstructions:

Obstruction 1: $\mathrm{sinc}(1) = 0$. Any perturbation $K \to K + \varepsilon\,\delta K$ gives $\delta Y_2(r) = 2\,\mathrm{sinc}(r)\,\delta K(r)$. At the typical GUE gap $r \approx 1$: $\mathrm{sinc}(1) = \sin(\pi)/\pi = 0$ exactly, so $\delta Y_2(1) = 0$ and the correction vanishes at the most important scale.

Obstruction 2: Sign. For any antisymmetric δK (which preserves $A_1[r] = 0$): $|K + i\varepsilon A|^2 \geq \mathrm{sinc}^2$ always, so Y_2 increases, triples become more unequal, and $\langle r \rangle$ decreases ($c < 0$). The physical correction requires $c > 0$.

Table 5: Summary of 29 perturbative approaches. All blocked by one of three obstructions.

#	Approach	c_{teo}	Obstruction
1	LeClair S-matrix	-0.009	quasi-lattice $\neq$ GUE
2	2nd order cosine A_2	0.004	$A_2 < 0$ for $\tau < 0.52$
3	Keating–Snaith factorisation	—	$\langle r \rangle$ not linear
4	Off-diagonal χ_2	-0.045	all $\chi_2 < 0$
5–6	Yakaboylu kernel	0.100	$K_{\text{Yak}} \neq$ GUE base
7	Resolvent 2nd order	400–1400	δK real-symmetric
8	Painlevé V BK	5–32	$Z_{\text{BK}} \neq 1$
9	DPP spectral	-38 to $+61$	$\sqrt{b_2} \neq K$
10	$\delta_2 R_3$ imaginary	$\sim 10^7$	$O(\sqrt{T})$ cancellation
11–14	Various density-preserving	$0.27\times$	$P_{\max} < T$ truncation
15–16	Line C (diagonal + ζ zeros)	e^{41}	non-perturbative
17	PV linearisation	310	$p_2 = p \times p$ fails
18	PV with exact p_2^{GUE}	1711	u_{dp} super-exponential
19–21	RH Szegő, L'Hôpital, Montgomery	0.086	PNT: 7% only
22–24	RH Lax pair, jump matrix	—	oscillatory $e^{2\pi i\lambda}$
25–27	Direct kernel, Born δY_2	26%	Born = max perturbative
28	Consolidation PNT+pairs	26%	$\delta Y_2(1) = -0.273 \neq 0$
29	GLM inverse	—	$Y_2^{\text{BK}}(1) < 0$, no DPP

Obstruction 3: $O(\sqrt{T})$ cancellation. The prime sum $f(r;T) = \sum_{p \le T} (\log p/\sqrt{p}) \sin(2\pi\tau_p r) \sim O(\sqrt{T}/\log T)$. The correction $c/\log^2 T$ emerges from perfect cancellation of $O(\sqrt{T})$ terms between the smooth (PNT) and oscillatory (zeros of ζ) parts. No numerical truncation at $P_{\max} < T$ captures this.

Resolution. The exact kernel $K^{\text{BK}} = \sqrt{\text{sinc}^2 + \varepsilon\,\delta Y_2}$ bypasses Obstruction 1 because $\sqrt{a^2+b}$ is smooth at $a = 0$. However, computing c from this kernel remains an open problem (Section 4 of main text). The ab initio route via Conrey–Snaith R_3 circumvents all three obstructions by working directly with the triple correlation [Paper B].

4 Supplementary Note 2: Proof details for Theorem 1

The complete rigorous proof is presented in Paper C ("Berry-Keating convergence implies the Riemann Hypothesis").

Here we provide the key intermediate results:

4.1 Lemma: $\|\delta K^{\text{exact}}\|_{\text{HS}} \sim \varepsilon^{3/4}$

For $\varepsilon > 0$ and δY_2 bounded:

$$\delta K(r) = \sqrt{\text{sinc}^2(r) + \varepsilon\,\delta Y_2(r)} - |\text{sinc}(r)|\,. \tag{1}$$

Near $r = 1$: $\text{sinc}(r) \approx -\pi(r-1)$, so $\text{sinc}^2(r) \approx \pi^2(r-1)^2$. For $|r-1| \ll \sqrt{\varepsilon}$: $\delta K \approx \sqrt{\varepsilon\,\delta Y_2(1)}$. For $|r-1| \gg \sqrt{\varepsilon}$: $\delta K \approx \varepsilon\,\delta Y_2(1)/(2\pi|r-1|)$.

The Hilbert–Schmidt norm:

$$\int_0^s |\delta K(r)|^2\,dr \sim \varepsilon^{3/2}\,, \quad \text{hence} \quad \|\delta K\|_{\text{HS}} \sim \varepsilon^{3/4}\,. \tag{2}$$

Numerically verified: ratio $\|\delta K\|_{\text{HS}}/\varepsilon^{3/4} \approx 2.8$, constant across $\varepsilon \in [10^{-5}, 10^{-1}]$.

4.2 Propagation bound

Simon's theorem gives $|\delta E(s)| \le \|\delta K_s\|_1\, e^{\|K_s^0\|_1 + \|K_s^{\text{BK}}\|_1 + 1} \le C(s)\,\varepsilon^{3/4}$ with $C(s) = e^{2s+1}$.

Since $E(s) = \det(I - K_s)$ is entire in s, Cauchy's integral formula with radius $h = 1$ gives:

$$|\delta p(s)| = |\delta E'(s)| \leq e^2\, C(s)\, \varepsilon^{3/4}\,. \tag{3}$$

The final bound:

$$|\delta\langle r\rangle| \leq \varepsilon^{3/4} \iint |r|\, e^2\, C(s_1{+}s_2)\, p_2^{\mathrm{GUE}}(s_1, s_2)\, ds_1\, ds_2 \approx 1182\, \varepsilon^{3/4}\,. \tag{4}$$

The integral converges because p_2^{GUE} decays as $e^{-O(S^2)}$ for $S = s_1 + s_2 \to \infty$, dominating $C(S) = e^{2S+1}$. Numerically: the integral stabilises at ≈ 1182 for $s_{\max} \geq 4$.